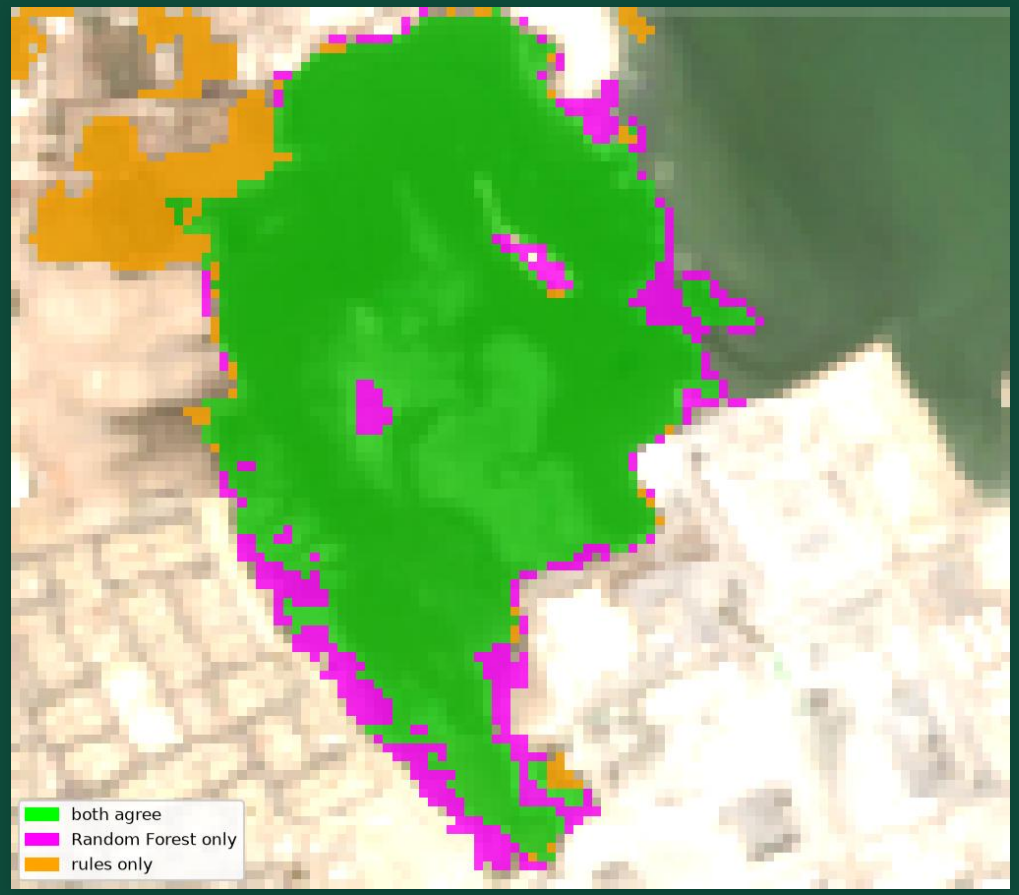
Bahrain Mangrove Monitoring

Mapping canopy from Sentinel-2 data with a Random Forest, and testing change against a noise floor



Sentinel-2 true color, 10 m pixels. Green = canopy mask.

Team Hmm

Arab Youth Space Hackathon: 813 Challenge

Theme: Ecosystem Health, Biodiversity & Blue Carbon

Track: Very High Resolution (VHR) | Country: Bahrain

Bahrain has lost more than 95% of its natural mangroves

328 → 48

hectares of natural mangrove, 1967 to 2020

more than 95% lost

Reclamation drove the loss

A University of Bahrain assessment attributes the decline mainly to land reclamation for urban development.

What remains is small and hard to watch

A stand of a few tens of hectares, squeezed between open water, tidal flats, farmland and industrial land.

Monitoring is periodic and manual

Public assessments digitize imagery by hand, leaving gaps between studies.

Sources: University of Bahrain assessment of the Tubli Bay area (digitalrepository.uob.edu.bh); Gulf Daily News.

Built for the people deciding where to look next



Who

Environmental authorities, conservation managers and mangrove restoration groups in Bahrain.



Decision

Where to send field teams, whether a stand is holding or losing canopy, and what baseline to restore against.



Today

Periodic studies that digitise satellite imagery by hand, such as the 1967 to 2020 University of Bahrain assessment.



What we add

A repeatable canopy map, a change test read against a noise floor, and a shortlist of places to check.

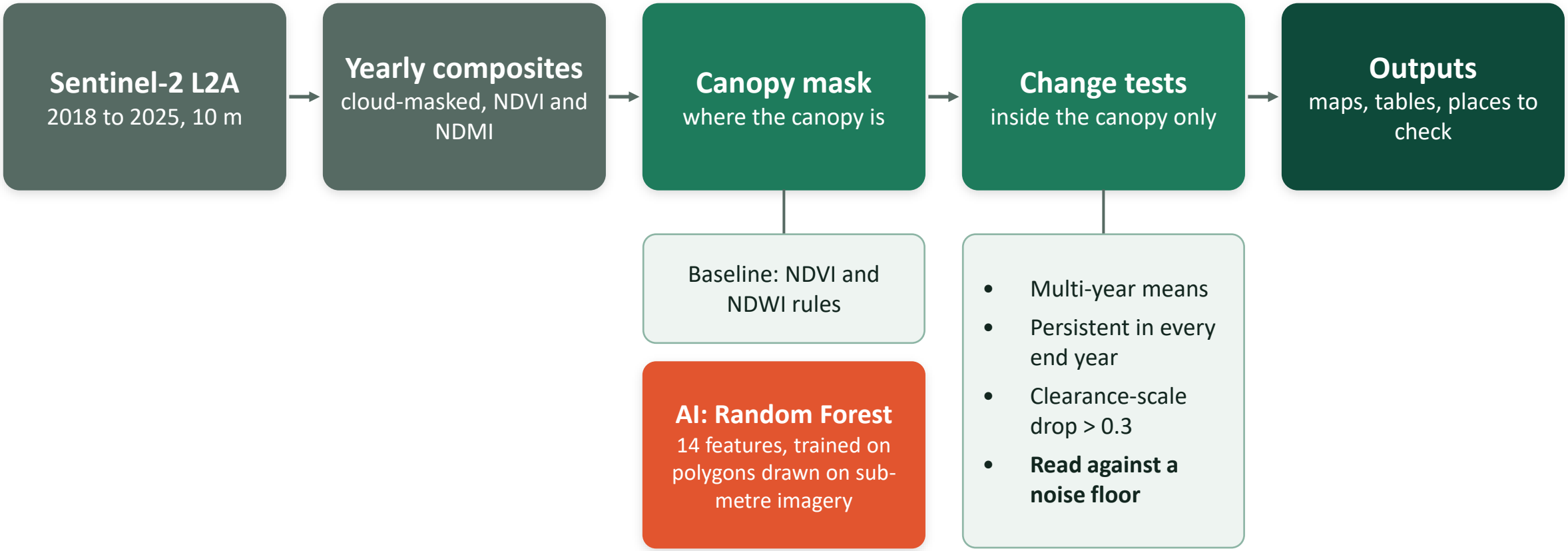
These users are hypothesised from public sources.

The right sensor for each job, and why

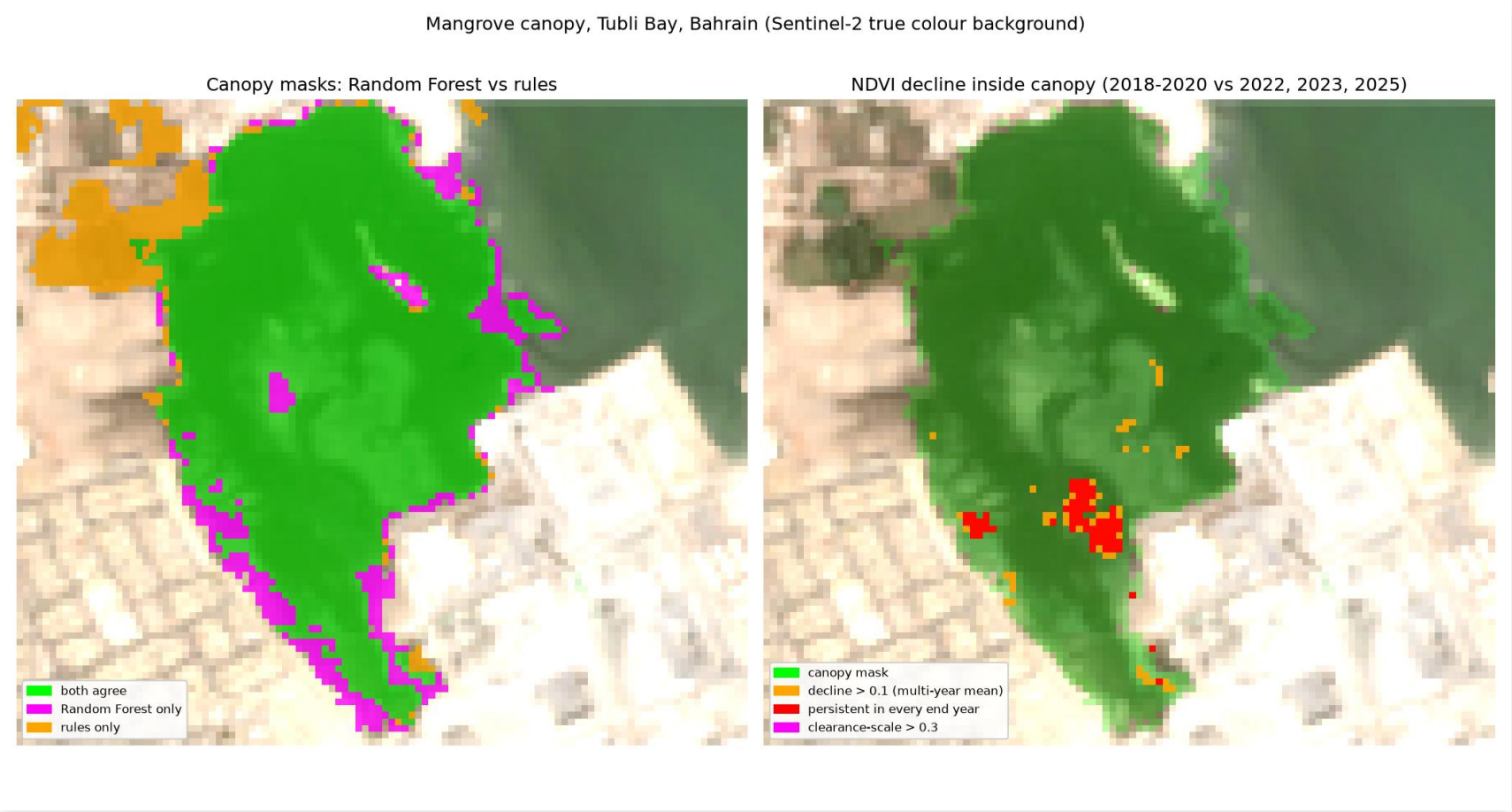
Source	Role in this PoC	Why	Status
Sentinel-2 **10 m, ~5-day revisit**	Analysis, 2018 to 2025	Archive covers the whole window and is frequent enough for yearly composites and a noise floor.	**Used**
Sub-metre imagery **(public basemap)**	Labels and visual checks	Detail to draw training polygons and inspect flagged patches. A handful of dates only, so not a time series.	**Used**
PlanetScope **3 to 4 m, daily**	Next step	A finer time series where sponsored access allows.	**Not used**

A 30 ha stand **≈ 330 pixels at 30 m** vs **≈ 3,000 at 10 m**

Map the canopy first, then measure change only inside it



Canopy mapped; persistent decline is concentrated in one southern patch



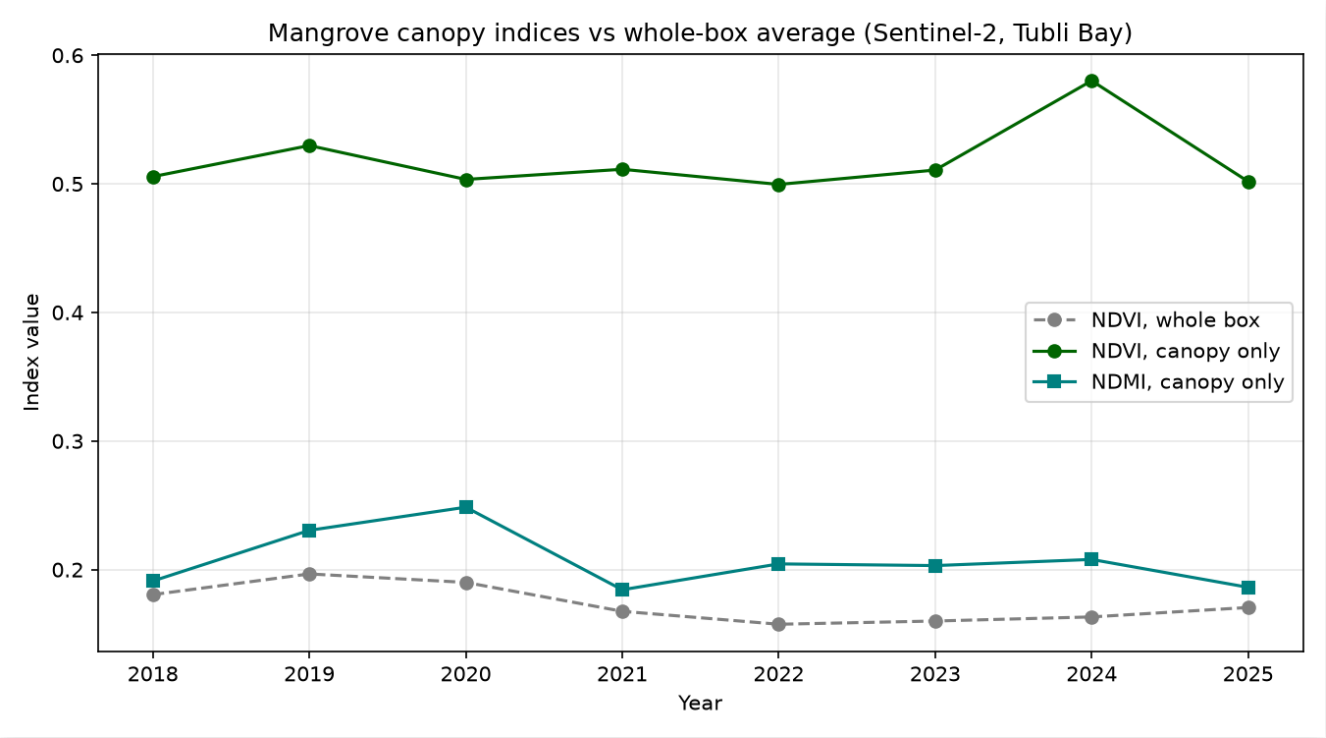
Left panel

Orange is rules only, mostly NW farmland and buildings. Magenta is model-only canopy edge.

Right panel

Red is below the baseline in every end year. One southern cluster is the place to check.

Change is judged against a noise floor, not in isolation



2024 is an outlier (canopy NDVI 0.58, against 0.50 to 0.53 in other years), so the end period leaves it out. The 2024 to 2025 pair flags 15.7 ha, which is that peak falling back, not loss.

Mean canopy NDVI, baseline vs end period

0.513 → 0.504

Decline > 0.1: end vs baseline, and the control period

1.44 ha vs 0.73 ha

Persistent clearance-scale loss (NDVI drop > 0.3)

0 ha

How far to trust the result

How we checked

- Whole polygons held out for testing, not pixels
- Rule-based mask scored on the same test pixels
- Change read against consecutive-year and control comparisons
- Flagged patches listed with coordinates for sub-metre checks

Canopy F1 on held-out polygons

Rule-based **0.895**

Random Forest **0.978**

Split: **pixel-level**

Labels are our own visual interpretation, not field data.

Limits

- No field data. Accuracy is agreement with our labels
- Test polygons sit inside homogeneous areas, so edges are under-tested
- 10 m pixels cannot see small clearings
- Tide, algae and odd years can mimic change

Repeatable monitoring from free data

Repeatable

The same notebook re-runs as new Sentinel-2 imagery arrives, with fixed seeds and documented parameters.

Targeted

Turns a coast-wide question into a short list of places to inspect, with coordinates.

Honest

Reports change beside a noise floor, so ordinary variability is not mistaken for loss.

Bahrain aims to quadruple its mangrove cover by 2035. Restoration only counts if the stands are watched.

Next steps for incubation

1

Confirm flagged patches on sub-metre imagery

Use sponsored high-resolution scenes to confirm or dismiss each flagged patch, and sharpen the canopy edges.

2

Remove the Earth Engine dependency

Rebuild on open STAC and cloud-optimised GeoTIFF sources so anyone can reproduce it without an account.

3

Add hyperspectral for stress and species

Use EnMAP, Tanager or Satellite 813 where spectral detail matters: early stress, salinity and species.

4

Validate in the field, then extend

Work with local agencies on field checks, then apply the method to other Bahrain stands.

Thank you

Bahrain Mangrove Monitoring | Team Hmm

Code and results: https://github.com/TheshadowM707/813-mangrove-monitoring

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